# Saifeddine Makhlouf

Lead Full-Stack Developer | React, Node.js, TypeScript | GenAI / LLM and Agentic Systems
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## PROFESSIONAL EXPERIENCE

**Lead Full-Stack Developer, YOTSYSTEM (Full-time)**
January 2026 - Present | Tunis, Tunisia (Hybrid) | AI-powered yacht management and digital transformation

- Build and own platform features across React, NestJS, TypeScript, and React Native in a 4-engineer team, running sprints and reviewing 100% of pull requests.
- Designed core backend architecture (NestJS main backend) across PostgreSQL and MongoDB handling 50,000+ API requests per day via REST and event-driven services.
- Built secure microservices, API gateways, and JWT/RBAC authentication flows powering production traffic; delivered the React.js back-office admin interface.
- Built Sonar AI, an agentic business-research back-office (React.js, MUI, LangGraph) that autonomously compiles complete business profiles (contacts, websites, insights), processing 1,000+ listings in bulk in under 15 minutes and replacing 15-30 minutes of manual research per business (99% time reduction).
- Shipped a cross-platform React Native and TypeScript mobile app with real-time WebSockets and Firebase against the same NestJS backend.
- Set up GitHub Actions CI/CD with Docker across staging and production, cutting release time by 60%; delivered 5+ major releases working directly with C-level stakeholders.

**Software Engineering Intern (PFE), CineBox (Part-time, concurrent)**
February 2026 - June 2026 | Tunisia | Cinema ticket booking platform (customer and admin)

- Built a responsive React.js front-end with reusable component architecture across customer booking and admin back-office flows (25 screens).
- Engineered a Spring Boot and Hibernate backend across 12 relational entities, sustaining 10,000 concurrent booking sessions in load tests.
- Implemented Apache Kafka event streaming for real-time seat-locking, achieving zero double-bookings across 10,000 concurrent sessions with confirmation latency under 100 ms.

**Freelance Full-Stack and GenAI Developer, Upwork / Fiverr**
April 2022 - December 2025 | Remote | 20+ production projects for international clients across web, mobile, and SaaS

- CraftGent: SaaS GenAI chatbot builder solving custom chatbot development costs for SMBs; multi-channel LLM platform (web widgets, SMS, Discord, Instagram, Facebook) with real-time analytics, Stripe billing, and 100+ customization options. Serves 1,000+ users across 700+ businesses, processing 100,000+ messages per month and automating an estimated 80% of routine customer queries. Stack: React, Node.js, Express, MongoDB, WebSockets, OpenAI GPT-4, Gemini.
- JuliaBot: Multi-agent GenAI platform replacing single-purpose chatbots with 7 specialized agents (Chat, Research, Medical, RAG), JWT RBAC, and 11-language i18n serving 500+ registered users. Stack: Next.js 15, NestJS, PostgreSQL, TypeORM, Docker, Kubernetes, Stripe.
- Calliverse: Social networking app with real-time messaging, video calls, and premium subscriptions. 1,000+ Google Play downloads with a 99%+ crash-free session rate. Stack: React Native, TypeScript, Node.js, Supabase, WebRTC, Clerk, Stripe.
- Blink Delivery: Grocery delivery app with live order tracking, multi-address delivery, coupon engine, and offline support. 2,000+ Google Play downloads, 5,000+ orders processed, 99%+ crash-free session rate. Stack: React Native, TypeScript, Node.js, WebSockets, Supabase, Stripe.
- Africa Creative Talent Expo: FinTech app with cashback rewards, proprietary voting currency, and real-time transaction processing. Stack: React Native, Node.js, Express, Supabase, Clerk, WebSockets.

**Software Engineering Intern, TOPNET**
January 2025 - February 2025 | Tunisia (Hybrid)

- Built a no-code chatbot builder platform from scratch using React.js, Node.js, and NLP APIs; redesigned NLP conversation flows, increasing queries resolved without human escalation by 25%.

## PERSONAL PROJECTS

**MagicScript**: From-scratch statically-typed, garbage-collected programming language (40,000+ lines of C++): one AST driving a tree-walking interpreter and an LLVM-IR compiler (clang, Boehm GC) with byte-identical output guaranteed across 1,141 tests. Generics, traits, null safety, green threads and channels, 13-module self-hosted standard library (http, json, sqlite, gfx3d), code formatter, Language Server (LSP), and VS Code extension.
Stack: C++17, LLVM, Boehm GC, CMake, GoogleTest

**RazenEngine**: Custom 3D game engine built from scratch with an ImGui editor and debug interface.
Stack: C++, OpenGL, ImGui

**Flowify**: AI prompt-to-workflow SaaS (n8n-style) removing the automation-tool learning curve: natural language compiled into executable workflows via LLM tool calling; 500+ workflows generated with roughly 90% valid on first generation.
Stack: Go, MongoDB, Next.js, LangChain, LangGraph

**TikAi**: Prompt-to-video AI editor: merges clips, auto-generates voice-overs and subtitles, and renders the final video from a single prompt, cutting 2-3 hours of manual editing to under 5 minutes end-to-end; 100+ videos rendered.
Stack: Python, FFmpeg, MoviePy, LangChain, LangGraph, ElevenLabs API, DeepSeek API

**EduAI**: AI learning platform (NotebookLM-style) preparing students for exams with flashcards, quizzes, and mock exams over an agentic RAG pipeline; 10,000+ study items (flashcards, quizzes) generated from user course material.
Stack: Python, React.js, LangChain, Google Gemini API

**FindPoint**: AI ATS recruiter platform: RAG and cosine-similarity embedding matching for precise CV-to-job filtering, ranking hundreds of CVs per opening in minutes instead of days of manual screening.
Stack: Vue.js, Python, LangChain

**Network Threat Classification Engine (IEEE AI Challenge, National Winner)**: Machine learning engine classifying network threats (DDoS and other attack classes) from incoming and outgoing packet features; 98% classification accuracy on public intrusion-detection datasets.
Stack: Python, supervised ML pipeline

**Dar (Play Store launch imminent)**: Airbnb-style rental marketplace for Tunisian locals and tourists with OTP phone authentication; launching with 100+ initial property listings.
Stack: React Native, Supabase, Firebase

**Win Mechi**: Carpooling app with real-time trip tracking on fully self-hosted mapping infrastructure (own Valhalla routing engine, open-source tile renderer and rasterizer server), reducing marginal routing cost to $0 versus roughly $5 per 1,000 requests on commercial Maps APIs.
Stack: React Native, WebSockets, Firebase, Hapi.js, Valhalla, OpenStreetMap

**Echri**: Shopify-like multi-vendor e-commerce platform: unlimited stores per platform with vendor-isolated inventory management and role-based admin dashboards.
Stack: React.js, Spring Boot

**RENTED**: Peer-to-peer rental marketplace for renting everyday items instead of buying them.
Stack: Laravel (PHP), Flutter

**Shrimp Time**: Restaurant website and landing page handling table bookings, reservations, and POS.
Stack: Remix (React)

**Khazna**: Point-of-sale and cash management desktop application (Gestion de Caisse) with local persistence; re-implemented in Python.
Stack: C, GTK, SQLite, Python

## SKILLS

**Frontend:** React.js, Next.js 15, Vue.js, Remix, TypeScript, Redux, Recoil, Tailwind CSS, MUI, Shadcn/UI
**Backend:** Node.js, Express.js, NestJS, Hapi.js, Go, Spring Boot (Java), Laravel (PHP), Python (FastAPI, Flask), REST, GraphQL, Microservices, WebSockets
**GenAI / LLM:** OpenAI GPT-4, Gemini, DeepSeek, LangChain, LangGraph, RAG, Multi-Agent Systems, Embeddings, Cosine Similarity, ElevenLabs, Conversational AI
**Systems:** C, C++17, LLVM, OpenGL, ImGui, GTK, Compilers, Interpreters, Language Servers (LSP)
**Databases:** PostgreSQL, MongoDB, MySQL, Redis, SQLite, Supabase, Prisma, TypeORM
**Mobile:** React Native, Expo, EAS Build, Flutter, WebRTC, Firebase, Push Notifications, Offline Sync
**DevOps and Cloud:** Docker, Kubernetes, GitHub Actions, GCP, AWS, Apache Kafka, Self-hosted Valhalla routing and tile servers
**Payments and Auth:** Stripe, Clerk, JWT, OAuth 2.0, RBAC
**Machine Learning:** Threat classification, supervised learning on network traffic datasets
**Programming Languages:** JavaScript, TypeScript, Python, Java, Go, PHP, C, C++

## EDUCATION

**Bachelor of Science in Computer Science, ISET Djerba (Higher Institute of Technological Studies)**
2023 - June 2026 | Tunisia | Graduated with Honors

## ACHIEVEMENTS AND CERTIFICATIONS

- National Winner, IEEE AI Challenge (Cybersecurity and Machine Learning)
- Google Cloud Platform Fundamentals; Google Kubernetes Engine; React Native (Udemy)

## LANGUAGES

English: C2 (Proficient) | French: C1 (Advanced) | Arabic: Native